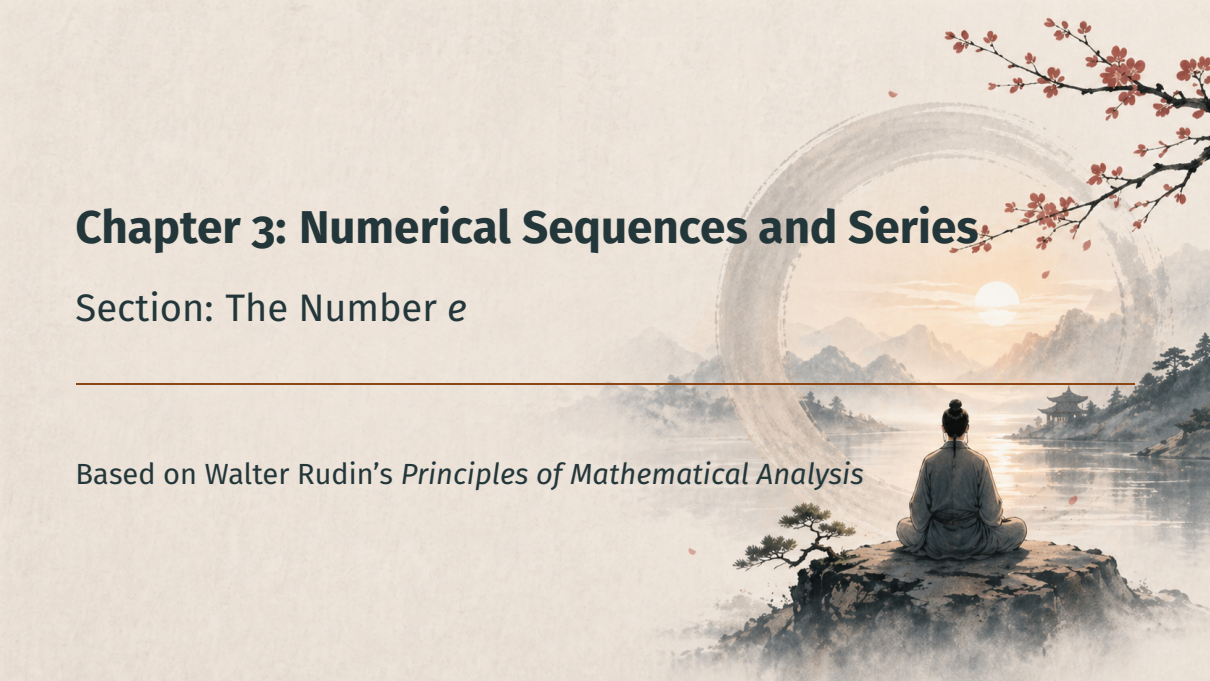


Chapter 3: Numerical Sequences and Series

The background is a traditional Chinese ink wash painting. A person with their hair in a bun, wearing a grey robe, sits in a meditative lotus position on a dark, craggy rock in the foreground. They are looking out over a calm lake that reflects the sky. In the distance, misty mountains rise. A large, faint, circular frame, resembling a ring or a large eye, is superimposed over the landscape, centered on the horizon. A small pavilion with a traditional roof is visible on the right side of the lake. Red cherry blossom branches with small flowers hang down from the top right corner. A few red petals are scattered in the air and on the water.

Section: The Number e

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

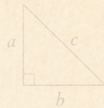
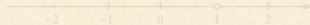
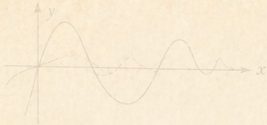
Defining the Number e

e as a Limit

How Fast the Series Converges

e is Irrational

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Why the Number e ?

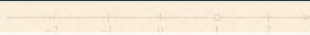
- e is one of the **fundamental constants** of mathematics



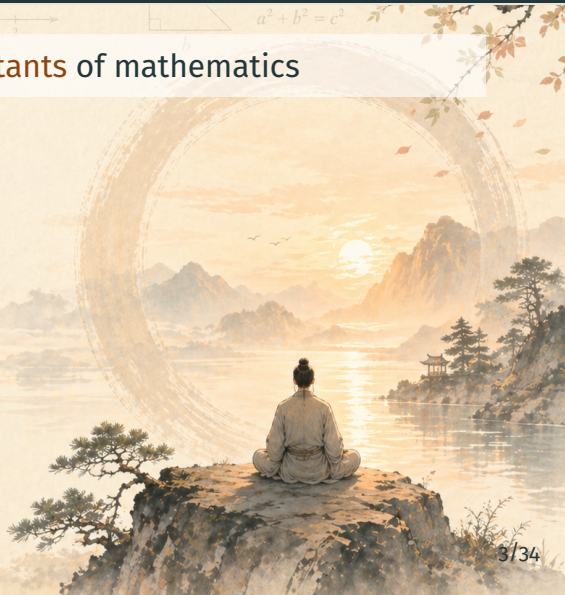
$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



$$a^2 + b^2 = c^2$$



Why the Number e ?

- e is one of the **fundamental constants** of mathematics
- We now have the tools to define it **rigorously**: series, limits



$$f(x) = \sin x$$

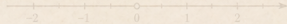


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Why the Number e ?

- e is one of the **fundamental constants** of mathematics
- We now have the tools to define it **rigorously**: series, limits
- Two surprises ahead:
 - e is also a **limit**: $\lim \left(1 + \frac{1}{n}\right)^n$
 - e is **irrational**

Defining the Number e



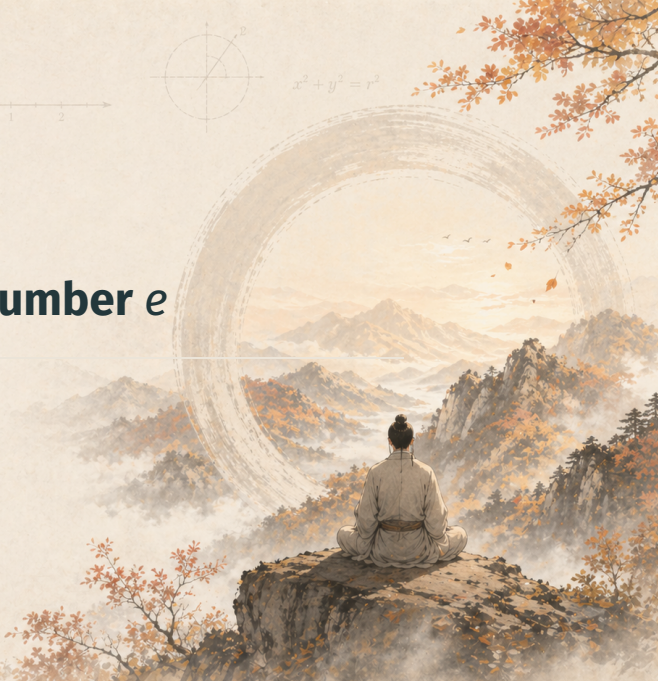
$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Definition 3.30: The Number e

Definition 3.30

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}.$$

Factorial convention:

$$n! = 1 \cdot 2 \cdot 3 \cdots n \quad (n \geq 1), \quad 0! = 1.$$

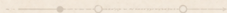
Does the Series Converge?

Write the n -th partial sum:

$$s_n = 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \cdots + \frac{1}{1 \cdot 2 \cdots n}.$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Does the Series Converge?

Write the n -th partial sum:

$$s_n = 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \cdots + \frac{1}{1 \cdot 2 \cdots n}.$$

Compare each term: for $k \geq 1$, $\frac{1}{k!} \leq \frac{1}{2^{k-1}}$.

$$s_n < 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}}.$$

Does the Series Converge?

Write the n -th partial sum:

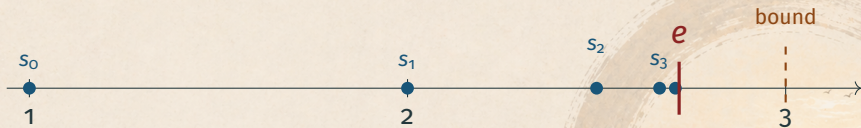
$$s_n = 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \cdots + \frac{1}{1 \cdot 2 \cdots n}.$$

Compare each term: for $k \geq 1$, $\frac{1}{k!} \leq \frac{1}{2^{k-1}}$.

$$s_n < 1 + \underbrace{1 + \frac{1}{2} + \cdots + \frac{1}{2^{n-1}}}_{< 2} < 3.$$

Bounded above by 3 $\implies$ the series converges.

The Partial Sums Close In on e



Increasing and bounded $\implies s_n \uparrow e$, with $2 < e < 3$.



$$a^2 + b^2 = c^2$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

A Series That Converges Very Fast

Remark

The series $\sum \frac{1}{n!}$ converges **very rapidly**.

- Each term shrinks faster than any geometric series.
- This lets us compute e with **great accuracy**.
- We will make “rapidly” precise with a sharp **error bound**.

e as a Limit

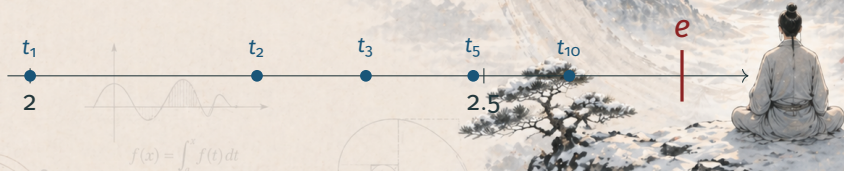


Theorem 3.31: e as a Limit

Theorem 3.31

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

Key idea: The number defined by a *series* also arises from a *limit process*.



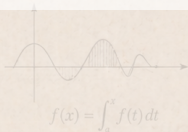
Even $t_{10} \approx 2.594$: this limit creeps toward e slowly.

Proof Strategy: Theorem 3.31

Goal: Show $t_n := \left(1 + \frac{1}{n}\right)^n \rightarrow e$.

Approach — squeeze between upper and lower limits:

1. Expand t_n with the **binomial theorem**.
2. Compare with s_n term by term: $t_n \leq s_n$, so $\limsup_{n \rightarrow \infty} t_n \leq e$.
3. Bound t_n from *below* by a truncated sum; deduce $e \leq \liminf_{n \rightarrow \infty} t_n$.
4. Combine: $\limsup t_n \leq e \leq \liminf t_n$ forces the limit.



Proof of Theorem 3.31 — Step 1: Setup

Step 1: Name the two sequences

$$s_n = \sum_{k=0}^n \frac{1}{k!}, \quad t_n = \left(1 + \frac{1}{n}\right)^n.$$

We already know $s_n \rightarrow e$. Goal: show $t_n \rightarrow e$ as well.

Proof of Theorem 3.31 — Step 2: Binomial Expansion

Step 2: Expand t_n by the binomial theorem

$$\begin{aligned} t_n &= 1 + \frac{n}{1!} \cdot \frac{1}{n} + \frac{n(n-1)}{2!} \cdot \frac{1}{n^2} + \cdots + \frac{n(n-1) \cdots 1}{n!} \cdot \frac{1}{n^n} = \\ &1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \frac{1}{3!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) + \cdots \\ &\quad + \frac{1}{n!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{n-1}{n}\right). \end{aligned}$$

Each term mirrors a term of s_n , but **dampened** by factors $\left(1 - \frac{j}{n}\right) \leq 1$.

Proof of Theorem 3.31 — Step 3: The Upper Limit

Since every factor $\left(1 - \frac{j}{n}\right) \leq 1$:

$$t_n \leq s_n \quad \text{for all } n.$$

Taking upper limits (Theorem 3.19):

$$\limsup_{n \rightarrow \infty} t_n \leq e. \quad (14)$$

Proof of Theorem 3.31 — Step 4: Truncate from Below

Strategy: keep only the first $m + 1$ terms (m fixed). For $n \geq m$,

$$t_n \geq 1 + 1 + \frac{1}{2!} \left(1 - \frac{1}{n}\right) + \cdots + \frac{1}{m!} \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{m-1}{n}\right).$$

All discarded terms are positive, so dropping them lowers the value.

Proof of Theorem 3.31 — Step 5: Let $n \rightarrow \infty$, m Fixed

In the truncated sum, let $n \rightarrow \infty$ with m fixed. Each factor $\left(1 - \frac{j}{n}\right) \rightarrow 1$:

$$\liminf_{n \rightarrow \infty} t_n \geq 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{m!} = s_m.$$

Hence $s_m \leq \liminf_{n \rightarrow \infty} t_n$ for every m .

Proof of Theorem 3.31 — Step 6: Let $m \rightarrow \infty$

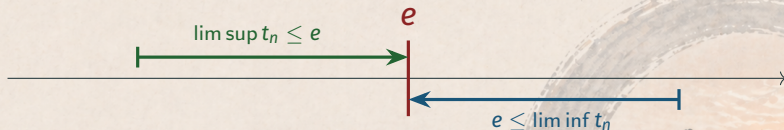
We have $s_m \leq \liminf_{n \rightarrow \infty} t_n$ for every m . Let $m \rightarrow \infty$: $s_m \rightarrow e$, so

$$e \leq \liminf_{n \rightarrow \infty} t_n. \quad (15)$$

Recall: Theorem 3.19

If $s_m \leq c$ for all m and $s_m \rightarrow e$, then $e \leq c$.

Proof of Theorem 3.31 — Conclusion

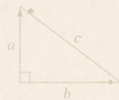
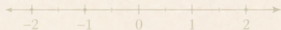


$$e \stackrel{(15)}{\leq} \liminf_{n \rightarrow \infty} t_n \leq \limsup_{n \rightarrow \infty} t_n \stackrel{(14)}{\leq} e.$$

All equal e , so

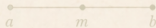
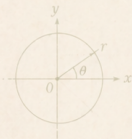
$$\boxed{\lim_{n \rightarrow \infty} t_n = e}$$





$$a^2 + b^2 = c^2$$

How Fast the Series Converges



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Estimating the Error $e - s_n$

How close is the partial sum s_n to e ? Look at the leftover tail:

$$e - s_n = \frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \frac{1}{(n+3)!} + \cdots$$

A Geometric Bound on the Tail

Factor out $\frac{1}{(n+1)!}$ and overestimate each factorial ratio:

$$e - s_n < \frac{1}{(n+1)!} \left\{ 1 + \frac{1}{n+1} + \frac{1}{(n+1)^2} + \cdots \right\} = \frac{1}{n! n}.$$

$$0 < e - s_n < \frac{1}{n! n}$$

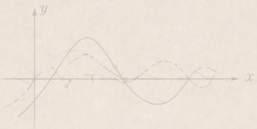
(16)

The Series Converges Astonishingly Fast

Example: $n = 10$

$$0 < e - s_{10} < \frac{1}{10! \cdot 10} = \frac{1}{36,288,000} < 10^{-7}.$$

- Just **11 terms** pin down e to better than 7 decimals.
- Estimate (16) also has **theoretical** value — see next section.

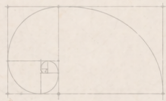


$$e^{i\theta} = \cos \theta + i \sin \theta$$



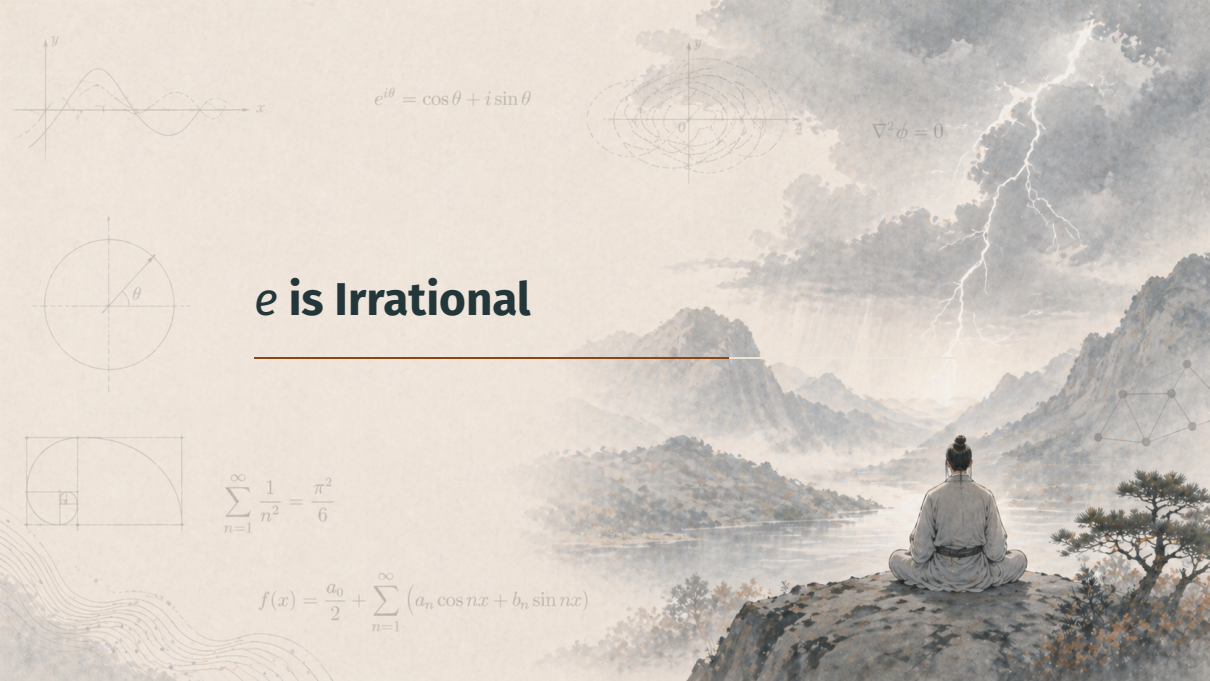
$$\nabla^2 \phi = 0$$

e is Irrational



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



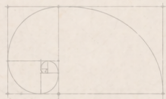
Theorem 3.32: e is Irrational

Theorem 3.32

e is irrational.

Meaning: there are *no* integers p, q with $e = \frac{p}{q}$.

Technique: proof by contradiction, powered by estimate (16).



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof Strategy: Theorem 3.32

Goal: rule out $e = p/q$ with p, q positive integers.

Approach:

1. Assume for contradiction $e = p/q$.
2. Multiply estimate (16) by $q!$: trap $q!(e - s_q)$ in $(0, 1)$.
3. Show $q!(e - s_q)$ is an **integer**.
4. An integer strictly between 0 and 1 — impossible!

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.32 — Step 1: Apply Estimate (16)

Step 1: Assume $e = p/q$ and scale (16) by $q!$

Suppose $e = p/q$ with p, q positive integers. From (16):

$$0 < q!(e - s_q) < \frac{1}{q}. \quad (17)$$

Since $q \geq 1$, the upper bound $\frac{1}{q} \leq 1$: $q!(e - s_q) \in (0, 1)$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.32 — Step 2: $q! e$ is an Integer

Under the assumption $e = p/q$:

$$q! e = q! \cdot \frac{p}{q} = (q-1)! p \in \mathbb{Z}.$$

The factor q in the denominator is cancelled by $q!$.

Proof of Theorem 3.32 — Step 3: $q! s_q$ is an Integer

$$q! s_q = q! \left(1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{q!} \right).$$

Each term $\frac{q!}{k!}$ with $0 \leq k \leq q$ is an integer $\implies q! s_q \in \mathbb{Z}$.

Proof of Theorem 3.32 — Step 4: The Contradiction

Both $q! e$ and $q! s_q$ are integers, hence so is their difference:

$$q! (e - s_q) = q! e - q! s_q \in \mathbb{Z}.$$

But (17) says $0 < q! (e - s_q) < 1$.

no integer lives here



An integer strictly between 0 and 1 — contradiction! ■

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Remark: e is Even Transcendental

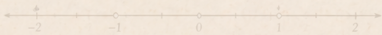
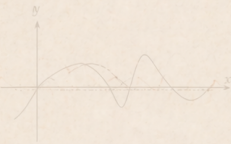
Remark

In fact e is not merely irrational — it is **transcendental**: it is *not* a root of any polynomial with integer coefficients.

- Irrational rules out $e = p/q$ (degree-one polynomials).
- Transcendental rules out *every* integer polynomial.
- Proof: see Niven, p. 25, or Herstein, p. 176.

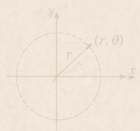
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



$$a^2 + b^2 = c^2$$

Summary



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways:

- **Definition 3.30:** $e = \sum_{n=0}^{\infty} \frac{1}{n!}$, with $2 < e < 3$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways:

- **Definition 3.30:** $e = \sum_{n=0}^{\infty} \frac{1}{n!}$, with $2 < e < 3$.
- **Theorem 3.31:** $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ by squeezing lim sup and lim inf.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways:

- **Definition 3.30:** $e = \sum_{n=0}^{\infty} \frac{1}{n!}$, with $2 < e < 3$.
- **Theorem 3.31:** $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ by squeezing lim sup and lim inf.
- **Estimate (16):** $0 < e - s_n < \frac{1}{n! n}$ with very fast convergence.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways:

- **Definition 3.30:** $e = \sum_{n=0}^{\infty} \frac{1}{n!}$, with $2 < e < 3$.
- **Theorem 3.31:** $e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ by squeezing lim sup and lim inf.
- **Estimate (16):** $0 < e - s_n < \frac{1}{n! n}$ with very fast convergence.
- **Theorem 3.32:** e is **irrational** (and in fact transcendental).

Coming up next: the *Root and Ratio Tests* — general tools for deciding convergence.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

